Corrigé — Exercice 12 • Bac principal 2024

1) $\det(A) = 1(1-2) - 0 + 1(6-23) = -1 - 17 = -18 \neq 0$: A inversible.

On calcule $A \times B = 18\, I_3$. Donc $A^{-1} = \dfrac{1}{18}B = \dfrac{1}{18}\begin{pmatrix} 1 & -2 & 1 \\ -20 & 22 & -2 \\ 17 & 2 & -1 \end{pmatrix}$.

2) (S) s'écrit $A\,X = \begin{pmatrix} 3 \\ 23 \\ 115 \end{pmatrix}$, d'où $X = A^{-1}\begin{pmatrix} 3 \\ 23 \\ 115 \end{pmatrix} = \dfrac{1}{18}B\begin{pmatrix} 3 \\ 23 \\ 115 \end{pmatrix} = \dfrac{1}{18}\begin{pmatrix} 72 \\ 216 \\ -18 \end{pmatrix} = \begin{pmatrix} 4 \\ 12 \\ -1 \end{pmatrix}$.

$$\boxed{(x, y, z) = (4,\ 12,\ -1)}$$